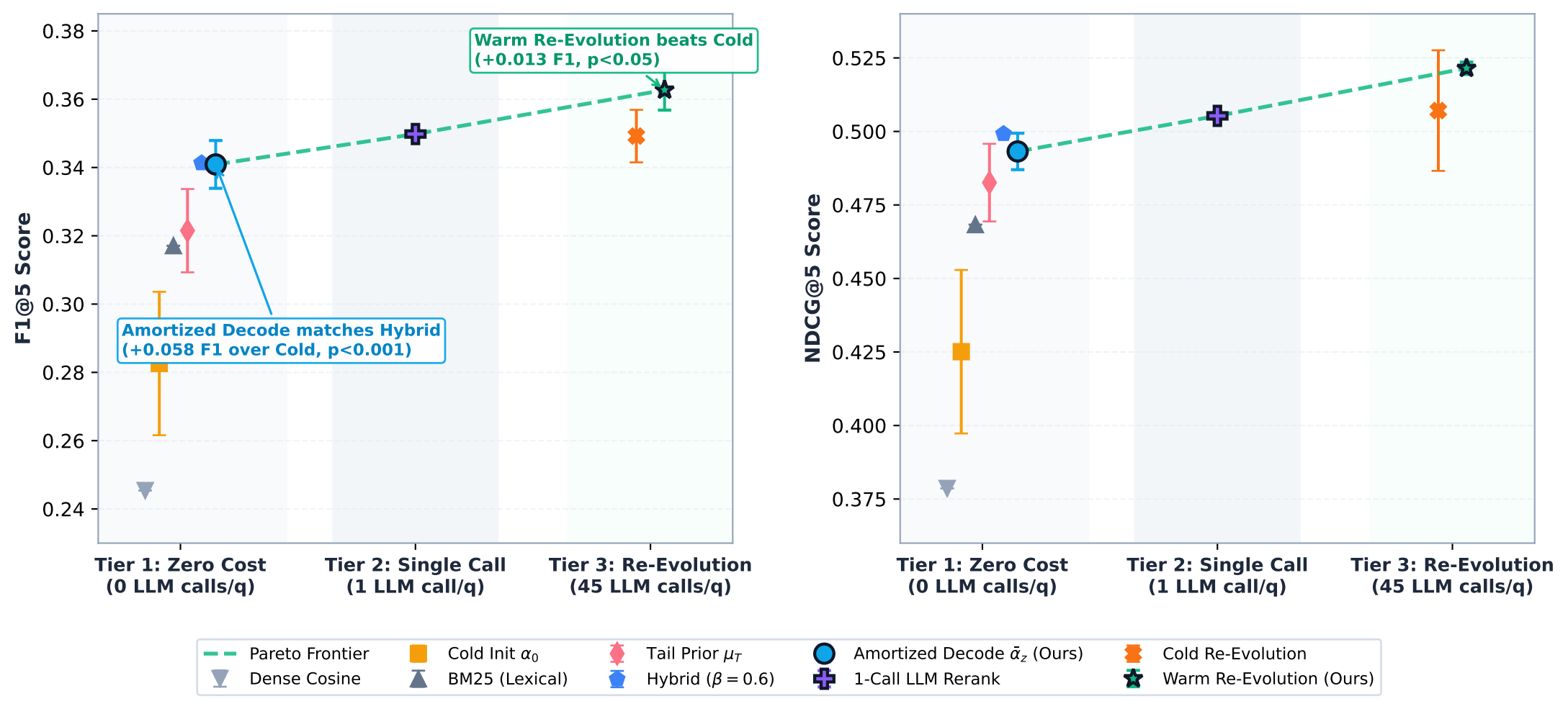
Warm Re-Evolution beats Cold
(+0.013 F1, p<0.05)
Amortized Decode matches Hybrid
(+0.058 F1 over Cold, p<0.001)
F1@5 Score
0.38
0.36
0.34
0.32
0.30
0.28
0.26
0.24
Tier 1: Zero Cost
(0 LLM calls/q)
Tier 2: Single Call
(1 LLM call/q)
Tier 3: Re-Evolution
(45 LLM calls/q)
NDCG@5 Score
0.525
0.500
0.475
0.450
0.425
0.400
0.375
Tier 1: Zero Cost
(0 LLM calls/q)
Tier 2: Single Call
(1 LLM call/q)
Tier 3: Re-Evolution
(45 LLM calls/q)
Pareto Frontier
Dense Cosine
Cold Init α_0
BM25 (Lexical)
Tail Prior μ_T
Hybrid ($\beta = 0.6$)
Amortized Decode $\bar{\alpha}_z$ (Ours)
1-Call LLM Rerank
Cold Re-Evolution
Warm Re-Evolution (Ours)